

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

B.Tech. EE Sem - VI Examination, May 2012
EE 307 Electrical Power System -III

Date: 08/05/2012, Tuesday Time: 01.30 p.m. to 04.30 p.m Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q – 1 Write the technical note on rating of circuit Breaker. 07
- Q – 2 A. Explain the arc interruption theories with their assumptions and limitations. 05
- B. Explain the different way due to which the ionization of gases occurs. 06
- C. Why kilometric fault is dangerous to Circuit Breaker? 03

OR

- Q – 2 A. Derive the expression for restriking voltage in terms of system capacitance and inductance. Also find the condition when the restriking voltage is maximum and RRRV is maximum. 08
- In a 220 KV system, the reactance and capacitance up to the location of C.B. is 06
- B. 8Ω and $0.025 \mu F$, respectively. A resistance of 600Ω is connected across the contacts of circuit breaker. Determine,
- (i) Natural frequency of oscillation
- (ii) Damped frequency of oscillation
- (iii) Critical value of resistance which will give no transient oscillation
- (iv) The value of resistance which will give damped frequency of oscillation, one fourth of the natural frequency of oscillation.
- Q – 3 A. Explain the function of 03
- (i) Arc runners
- (ii) Arc splitters
- (iii) Magnetic blow out coil.
- B. Explain any one method for artificial current zero in Direct Current Circuit Breaker. 04
- C. With neat sketch explain the construction and working of Air Break Circuit Breaker. 07

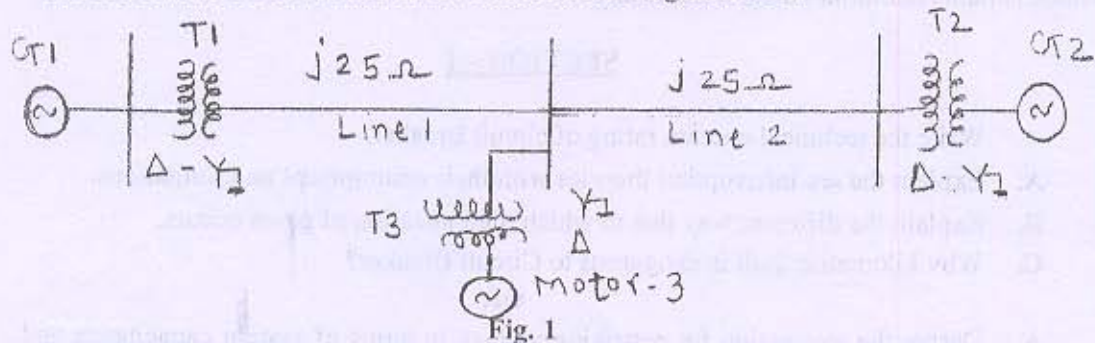
OR

- Q – 3 A. Explain the construction and working of single pressure puffer type SF_6 Circuit Breaker. 07
- B. Explain the construction and working of Vacuum Circuit Breaker. 07

SECTION - II

- Q-4 State the advantages of per unit system. Draw the per unit reactance diagram for the system shown in Fig. 1. 07

Generator 1: 25 MVA, 11 kV, $X_d = j 0.2$ pu
 Generator 2: 15 MVA, 11 kV, $X_d = j 0.2$ pu
 Motor 3: 25 MVA, 11 kV, $X_d = j 0.2$ pu
 Transformer 1: 25 MVA, 11 Δ / 120 Y kV, $X = j 0.1$ pu
 Transformer 2: 12.5 MVA, 11 Δ / 120 Y kV, $X = j 0.2$ pu
 Transformer 3: 10 MVA, 120 Y / 11 Y kV, $X = j 0.15$ pu
 Line 1 and Line 2: $X = j 25$ ohms
 Choose base of 50 MVA and 11 kV in the circuit of generator 1.



- Q-5 A. Derive the expression for the fault current using a simple R - L series circuit. 04
 State the two extreme conditions for the presence of DC offset component in the fault current. Also draw the waveform of the fault current for the following conditions: (a) DC offset current present (b) DC offset current absent
 B. Compute the following: (1) $a^2 - 1$ (2) $1 - a - a^2$ (3) $3a^2 + 4a + 2$ 03
 C. For the system shown in Fig. 2, each of the generators G1 and G2 rated at 125 MVA, 11 kV and has a sub transient reactance of 0.21 pu. Each of the transformers are rated at 125 MVA, 11 kV / 132 kV and has a leakage reactance of 0.06 pu. Find fault current and fault MVA for a fault at bus 5. 07

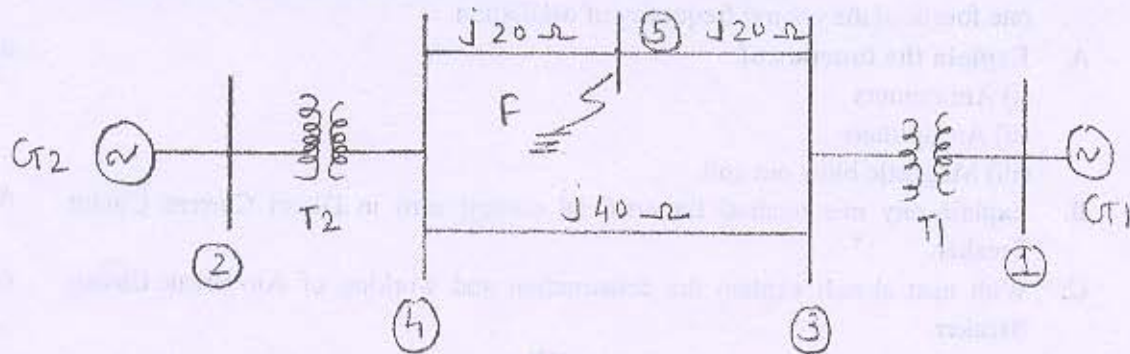


Fig. 2

OR

- Q-5 A. Derive the whole expressions for drawing the positive, negative and zero sequence networks for the synchronous generator. Also draw three sequence networks for the synchronous generator. 07

- B. A delta connected balanced resistive load is connected across an unbalanced three phase supply as shown in Fig. 3. With the currents in lines A and B specified, find the symmetrical components of line currents. Also find the symmetrical components of delta currents. Give your comments on the relationship between symmetrical components of line and delta currents. 07

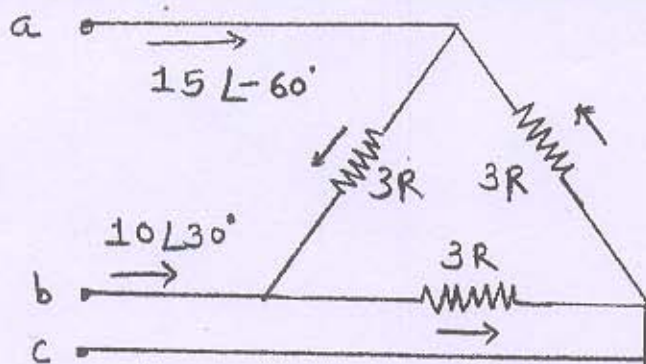


Fig. 3

- Q-6 A. A synchronous generator is grounded through the reactance X_n . The generator has sequence reactance X^1, X^2 and X^0 such that $X^1 = X^2 \gg X^0$. 07
- Draw the sequence networks of the generator as seen from the terminals
 - Show the expression for the fault current for a solid line-to-ground fault on phase a.
 - Show that, if a neutral is grounded solidly, the LG fault current would be more than three-phase fault
 - Write the expression for the neutral grounding reactance, such that the LG fault current is less than three phase fault current
- B. A 30 MVA, 11 kV generator has $Z^1 = Z^2 = j 0.2$ pu, $Z^0 = j 0.05$ pu. (a) A single line to ground fault occurs at the generator terminals. Find the fault current and line to line voltages during the fault conditions. Also, find the fault the fault current, when three phase fault occurs. 07

OR

- Q-6 A. Derive the expressions for the fault current for Line-to-Line (L-L) fault and draw the sequence network in the event of L-L fault. 07
- B. For the system shown in Fig. 4, a double line to ground fault occurs at the bus 2. Find the line currents under the fault conditions. 07

